

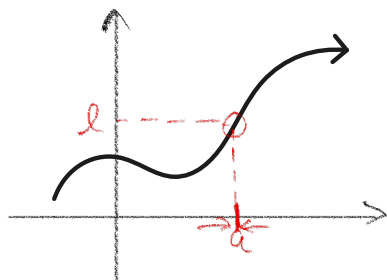
V. Limits

Def. The function $f: D \rightarrow \mathbb{R}$ approaches the limit l near a if for every $\varepsilon > 0$, $\exists$ some $\delta > 0$.

s.t. $\forall x$ with $0 < |x - a| < \delta$

then we have $|f(x) - l| < \varepsilon$

$$\hookrightarrow \boxed{\lim_{x \rightarrow a} f(x) = l}$$



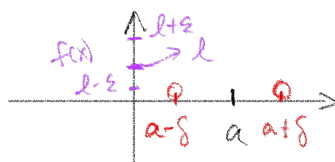
• What does $0 < |x - a| < \delta$ mean?

$$0 < |x - a| < \delta$$

$$\Leftrightarrow |x - a| < \delta, \text{ and } x \neq a.$$

$$\Leftrightarrow -\delta < x - a < \delta, x \neq a$$

$$\Leftrightarrow a - \delta < x < a + \delta, x \neq a$$



• What does $|f(x) - l| < \varepsilon$ mean?

$$\Leftrightarrow -\varepsilon < f(x) - l < \varepsilon$$

$$\Leftrightarrow l - \varepsilon < f(x) < l + \varepsilon.$$

Note • I give you ε . So you have no control over it.

• Your job is to find $\delta > 0$ s.t
whenever x is in $(a - \delta, a + \delta)$.

$f(x)$ is fixed to be in $(l - \varepsilon, l + \varepsilon)$.

Ex. Let $f(x) = 2x$. Show $\lim_{x \rightarrow 0} f(x) = 0$.

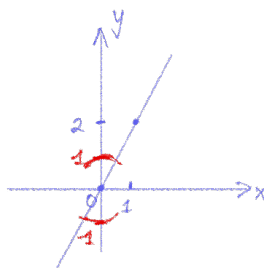
Roughwork

What if $\varepsilon = 1$?

$$\hookrightarrow \text{If } -\frac{1}{2} < x < \frac{1}{2}$$

$$\text{then } -1 < f(x) < 1$$

So take $\delta = \frac{1}{2}$ (or anything smaller)



What if $\varepsilon = 1/2$?

$$\hookrightarrow \text{If } -1/4 < x < 1/4$$

$$\text{then } -1/2 < f(x) < 1/2$$

So take $\delta = \frac{1}{4}$.

$\Rightarrow$ Guess $\delta = \frac{\varepsilon}{2}$ works.

PF. Let $\varepsilon > 0$. Want to find $\delta > 0$ s.t. whenever
 $0 < |x - 0| < \delta$

We can get $|f(x) - 0| < \varepsilon$.

* If we can do this, we're shown

$\lim_{x \rightarrow 0} f(x) = 0$ by definition.

We claim that $\delta = \frac{\epsilon}{2}$ works.

That is whenever $0 < |x-0| < \epsilon/2$,

$$|f(x)-0| < \epsilon.$$

Since we're given that $0 < |x| < \frac{\epsilon}{2}$

$$\Rightarrow 0 < |2x| < \epsilon$$

$$\Rightarrow |2x| < \epsilon.$$



Ex. let $f(x) = 2x$. Show that $\lim_{x \rightarrow 2} f(x) = 4$.

$$\epsilon = 1 \rightarrow \delta = \frac{1}{2}$$

$$\epsilon = 1/10 \rightarrow \delta = 1/20$$

$$\Rightarrow \boxed{\delta = \frac{\epsilon}{2}} \text{ Guess}$$

Pf. let $\epsilon > 0$. Want to find $\delta > 0$.

s.t. whenever

$$0 < |x-2| < \delta$$

then we have $|f(x)-4| < \epsilon$.

We claim $\delta = \epsilon/3$ works.

Whenever $0 < |x-2| < \frac{\epsilon}{3}$

we have $|2x-4| < \epsilon$.

Given $0 < |x-2| < \epsilon$.

Multiply by 2. $0 < 2|x-2| < 2\epsilon/3$.

$$\Downarrow \\ |2x-4| < \frac{2\epsilon}{3} < \epsilon$$

So the δ that we've chosen works.



Ex. Let $f(x) = x^2$. Show $\lim_{x \rightarrow 3} f(x) = 9$

Pf. Let $\varepsilon > 0$. Want $\delta > 0$, s.t.,

whenever $0 < |x-3| < \delta$

we have $|x^2 - 9| < \varepsilon$.

Note $|x^2 - 9| = |x-3||x+3| < \delta|x+3|$.

Assume $\delta < 1$ always.

Then $5 < |x+3| < 7$

b/c, $|x-3| < \delta$

$\Rightarrow |x^2 - 9| < \delta|x+3| = 7\delta$

Thus $\delta < \min\{\varepsilon/7, 1\}$.

